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$$\lim_{x \to \frac{\pi}{2}} \cos 5x \cot 4x$$

Stel $u = x - \dfrac{\pi}{2}$

$$\cos 5x = \cos 5\left(\frac{\pi}{2} + u\right) = \cos\left(\frac{5\pi}{2} + 5u\right) = -\sin 5u$$

$$\cot 4x = \cot 4\left(\frac{\pi}{2} + u\right) = \cot\left(2\pi + 4u\right) = \cot 4u = \frac{\cos 4u}{\sin 4u}$$

$$= \lim_{u \to 0}\left(-\sin 5u \cdot \frac{\cos 4u}{\sin 4u}\right)$$

$$= \lim_{u \to 0}\left(-\frac{\sin 5u}{5u} \cdot \frac{4u}{\sin 4u} \cdot \cos 4u \cdot \frac{5u}{4u}\right)$$

$$= -1 \cdot 1 \cdot 1 \cdot \frac{5}{4}$$

$$= -\frac{5}{4}$$

$$\Rightarrow \lim_{x \to \frac{\pi}{2}} \cos 5x \cot 4x = -\frac{5}{4}$$

References